

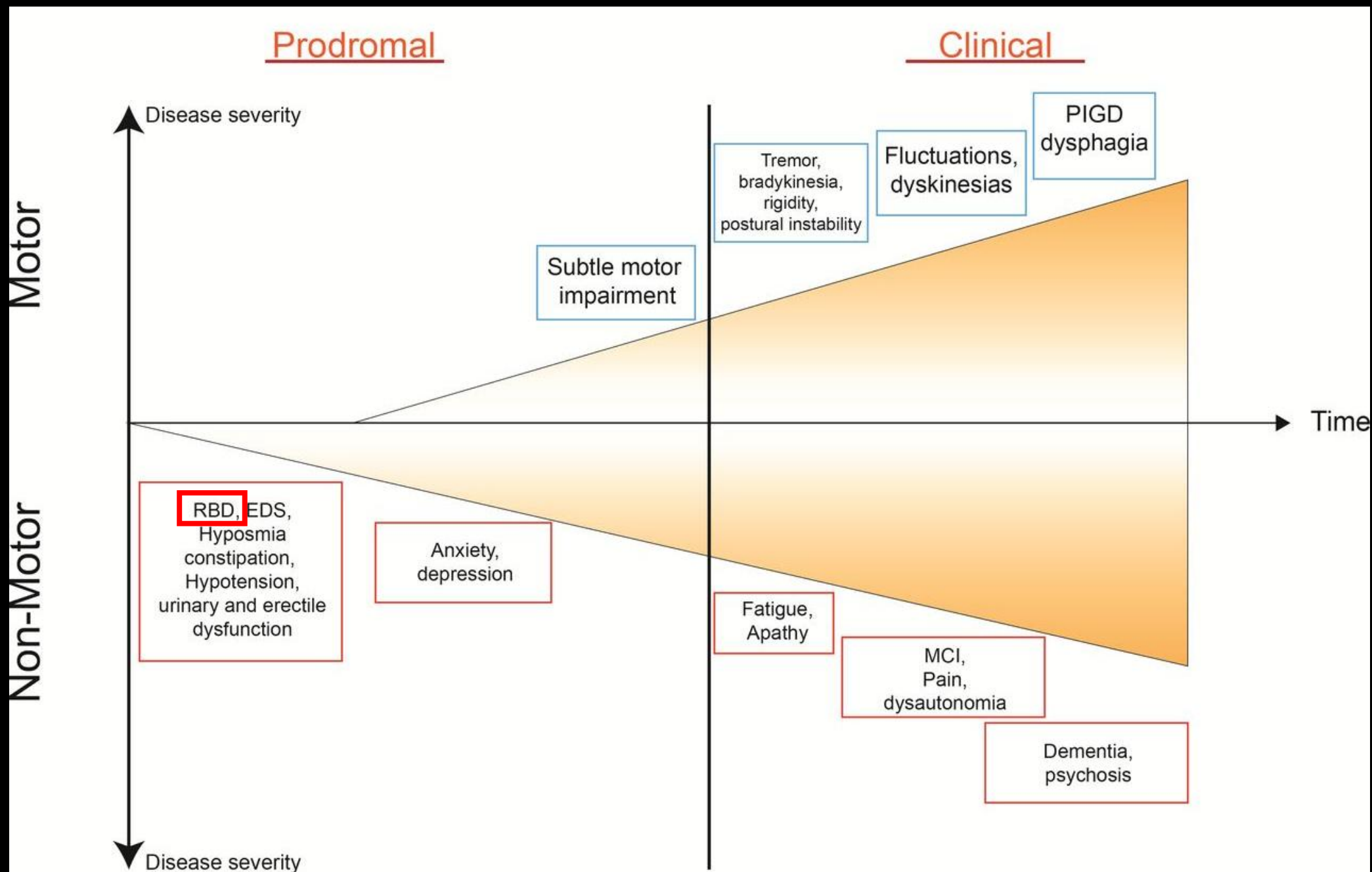
From PSG Signals to Deep Learning - An AI-Assisted Hands-on Approach -

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What this workshop session is

- A walkthrough of the most basic deep learning workflow
- Experience how AI-assisted coding changes what's possible for us
- Not a Python crash course
- Not a deep learning crash course



American Academy of Sleep Medicine

- Repeated episodes of sleep-related vocalization, complex motor behaviors, or both.
 - Repetitive episodes during a single night of video-polysomnography.
- Behaviors occur during REM sleep.
 - Shown by polysomnography, or are presumed to occur during REM sleep on the basis of clinical history of dream enactment.
- Polysomnographic recording shows REM sleep without atonia.
- The disturbance is not better explained by another sleep disorder, mental disorder, drug, or substance use.

Goal

- Lets try build a deep learning model detecting REM sleep without atonia (RSWA) in polysomnography.
 - To make things easier, the question we explore is “classification between normal REM and RSWA”.
- Let’s use it to annotate REM sleep behavior disorder in patients.

Outline

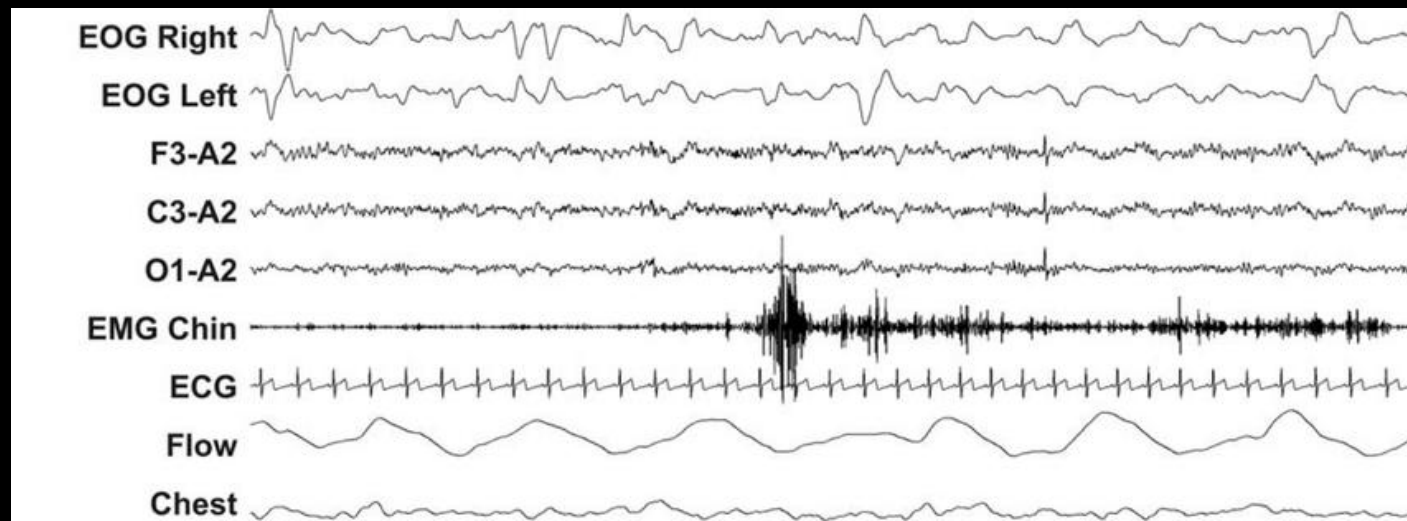
- Module 1 · From PSG signals to training data
- Module 2 · Teaching a 1D-CNN to detect RSWA
- Module 3 · Fighting overfitting
- Module 4 · Opening the black box

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PhysioNet CAP Sleep Database

	RBD	control
Original data	22	16
Exclude	-1 (Duplicated data; rbd10, rbd11)	-4 (Low signal sampling rate; n6, n7, n9, n12) -1 (No EMG channel; n16) -3 (Broken EMG signal; n13, n14, n15)
Final patients	21	8



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History

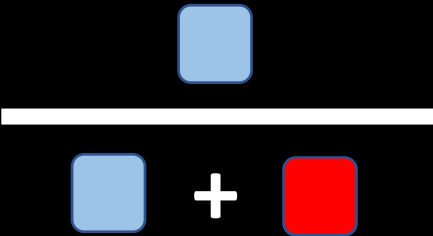
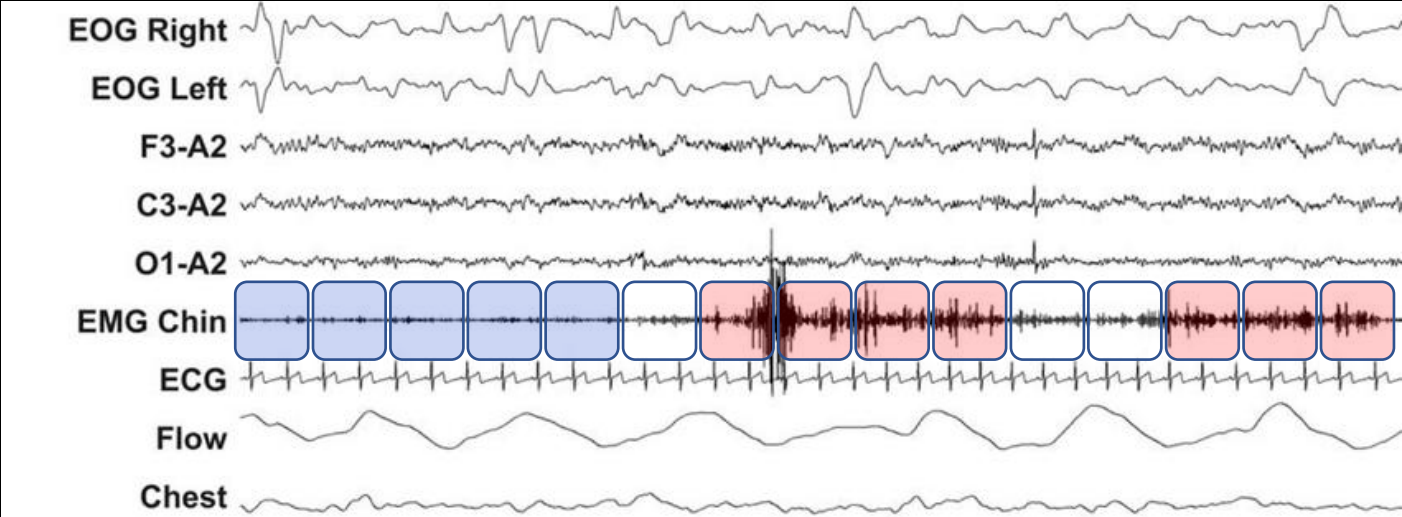
Name	Last commit message	Last commit date
..		
README.md	Upload notebooks	11 hours ago
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make_demo_clips.py	Upload notebooks	11 hours ago
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README.md

Data preparation scripts

A slim, self-contained pipeline that regenerates the workshop dataset from the public **PhysioNet CAP Sleep Database** — provided for transparency and reproducibility. Participants do not need these: the notebook downloads the prepared data from the GitHub Release automatically.

REM sleep without atonia



Behav Neurol. 2010;22(3-4):91-9.

	AASM 2023	Our label	Ferri REM atonia index
Epoch duration	Tonic: 30 sec Phasic: 3 sec	3 sec	1 sec
Amplitude	Tonic: 2X NREM background Phasic: 4x NREM background	Tonic: 2 μ V Phasic: 3 μ V	≤ 1 , 1–2, $> 2\mu$ V
RBD Threshold	Not clearly defined	$> 18\%$	RAI < 0.8 $p(\leq 1\mu V) / (1 - p(1-2\mu V))$

Confusion Matrix (Type I and II Error):

		Condition (as determined by "Gold standard")		
		Condition positive	Condition negative	
Test outcome	Test outcome positive	True positive	False positive (Type I error)	Precision = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Test outcome positive}}$
	Test outcome negative	False negative (Type II error)	True negative	Negative predictive value = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Test outcome negative}}$
		Sensitivity = $\frac{\Sigma \text{ True positive}}{\Sigma \text{ Condition positive}}$	Specificity = $\frac{\Sigma \text{ True negative}}{\Sigma \text{ Condition negative}}$	Accuracy

Precision:

$$PPV = \frac{TP}{TP + FP}$$

Negative Predictive Value:

$$NPV = \frac{TN}{TN + FN}$$

Recall (Sensitivity):

$$Sensitivity = \frac{TP}{TP + FN}$$

Specificity:

$$Specificity = \frac{TN}{TN + FP}$$

F1 Score:

$$F1 \text{ Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

1 = Perfect

0 = Poor

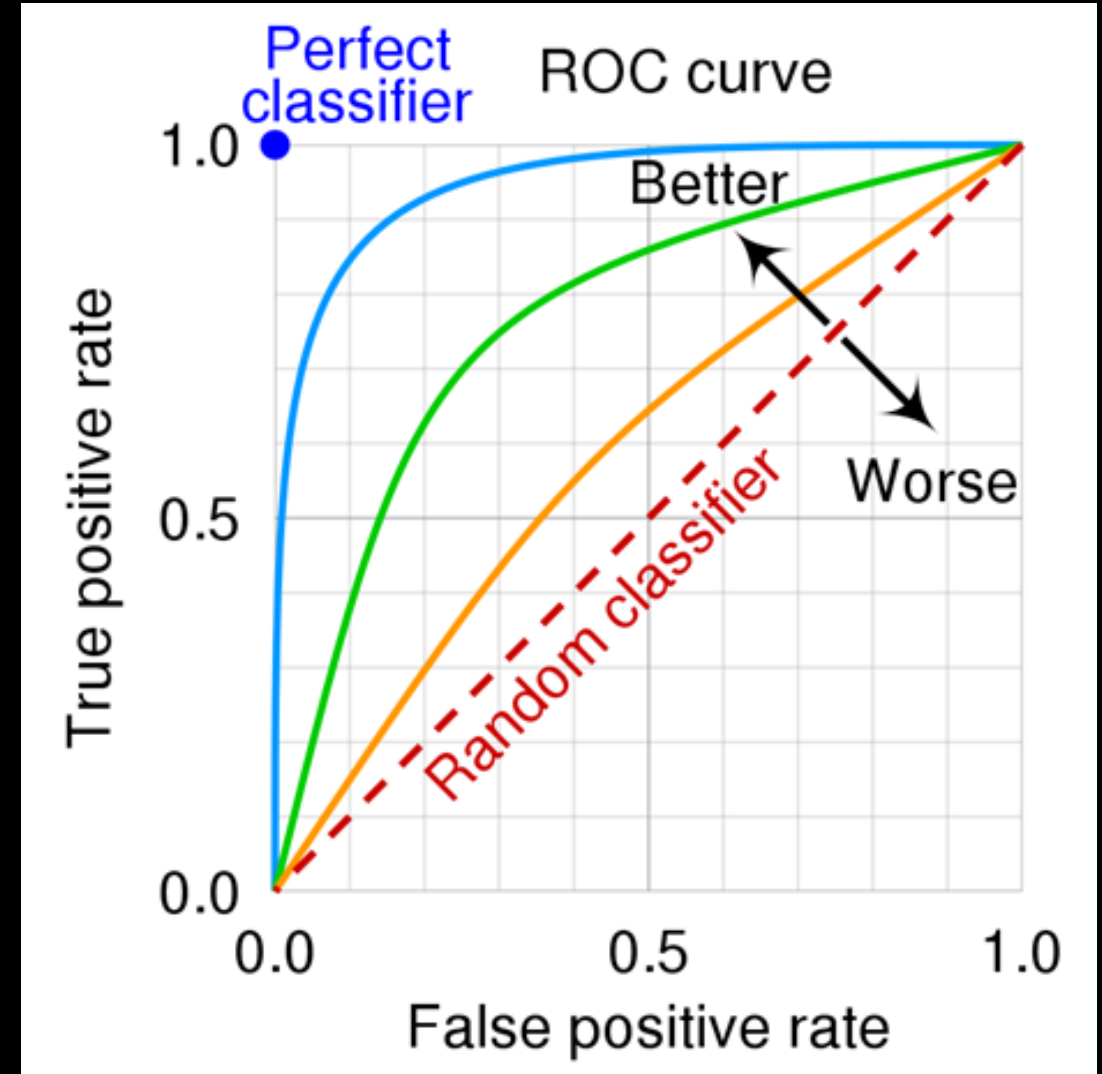
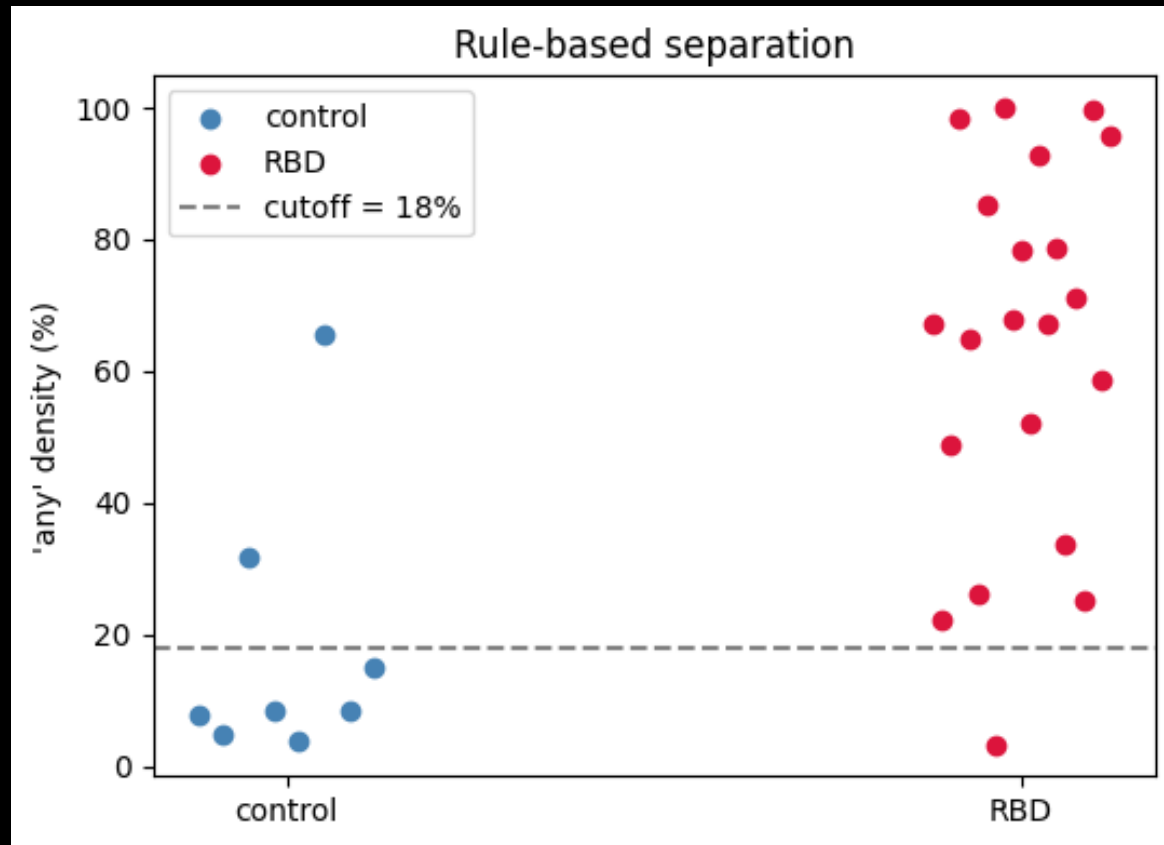
False Negative Rate:

$$FNR = \frac{FN}{FN + TP}$$

False Positive Rate:

$$FPR = \frac{FP}{FP + TN}$$

Receiver Operating Characteristic



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Releases 1

 **Workshop data v1 (RSWA)** Latest

2 weeks ago

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AOC-IFCN Workshop: Generative AI for Sleep Signals

through Vibe Coding

https://github.com/Po-yuLin/AOC-IFCN_workshop



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Text Editor



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XML, JSON Viewer



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Platform we use

The screenshot displays the Google Colab environment within the Visual Studio Code editor. The interface is divided into several sections:

- Top Bar:** Includes the Colab logo, a "Welcome To Colab" message with a warning "Cannot save changes", and a menu bar with options: File, Edit, View, Insert, Runtime, Tools, and Help. On the right, there are icons for settings, a "Share" button, and a user profile icon labeled "P".
- Left Sidebar:** Contains a "Table of contents" panel with a list of links: "Welcome to Colab!", "Google Colab is available in V...", "Access popular AI model...", "Explore the Gemini API", "Getting started", "Data science", "Machine learning", "More Resources", and "Featured examples". Below this is a "+ Section" button.
- Main Editor Area:** Displays a "Welcome to Colab!" message. It states "Google Colab is available in VS Code!" and provides instructions on how to use the Google Colab extension for Visual Studio Code. The instructions include: opening the **Extensions** view, searching for 'Google Colab', installing it, opening the kernel selector by creating or opening any **.ipynb** notebook file, running a cell or clicking the **Select Kernel** button, and clicking **Colab** to select a runtime and sign in with a Google account. A code cell at the bottom contains the following Python code:

```
from google.colab import ai
response = ai.generate_text("What is the capital of France?")
```
- Right Panel:** Labeled "Gemini", it shows a chat interface with the greeting "Hello, Po-Yu" and the question "How can I help you today?". Below the chat are three buttons: "How do I upload a dataset to this notebook?", "Show me how to install and import common data science libraries", and "Write a code snippet to print 'Hello, Colab!'". At the bottom of the panel, there is an "Empty cell" button and a prompt "What can I help you build?". The bottom right corner shows "Gemini 2.5 Flash" with a dropdown arrow and a play button.

Large language model (LLM) assisted coding

- Generate from scratch: state your goal clear.
- Modify existing code: what you see, what you want to modify.
- Want to understand the code: ask LLM to explain.
- Encountering bug: paste the code, bug information (including traceback) to the LLM.

Outline

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A neuron (perceptron) in artificial neural network

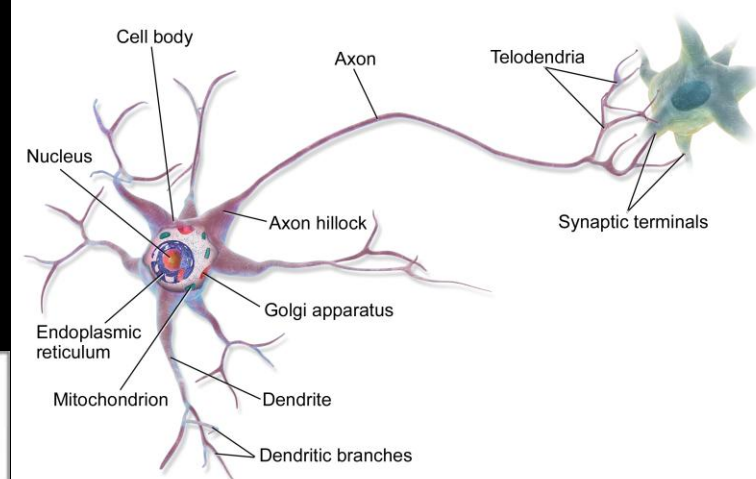
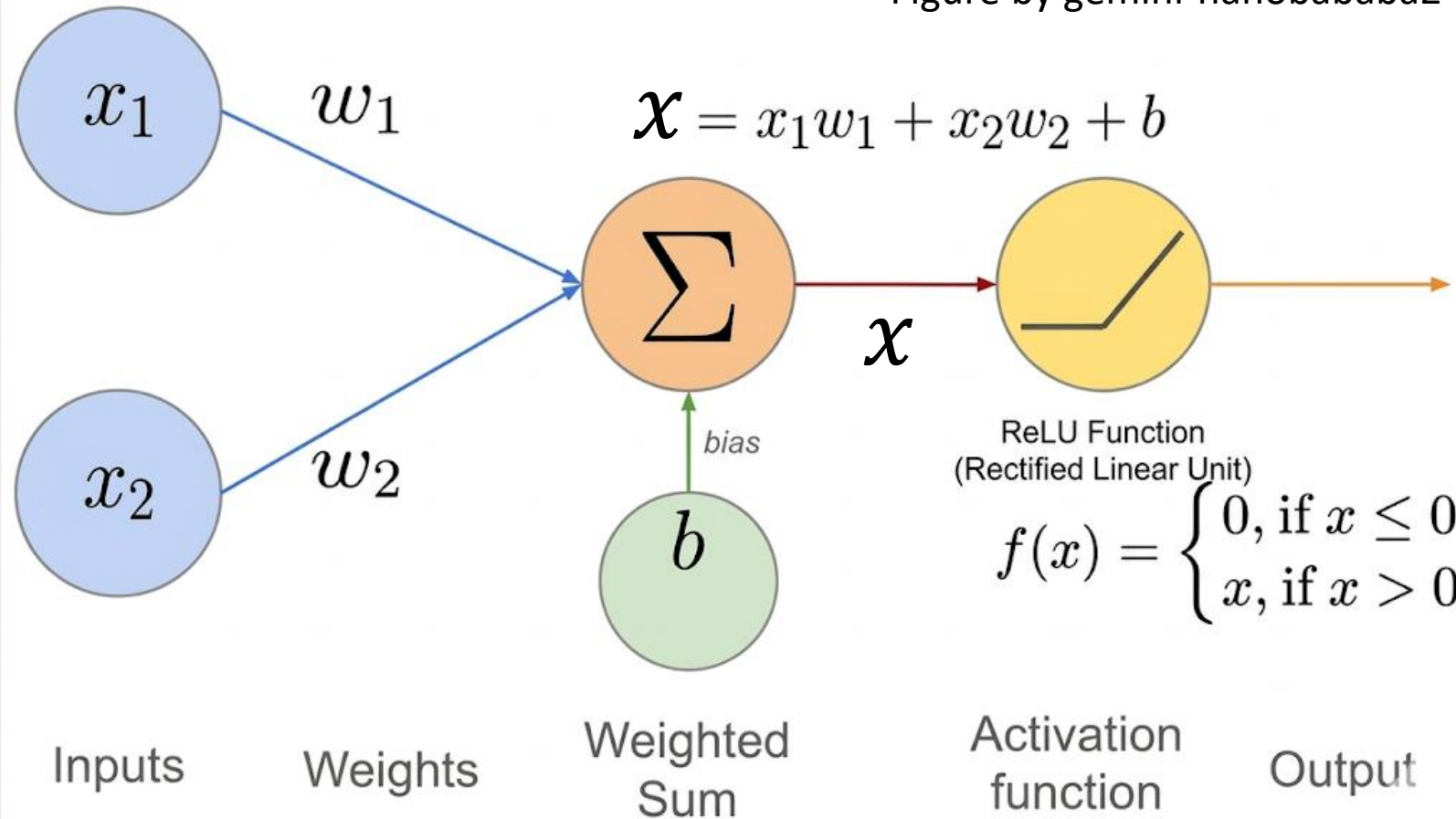


Figure by gemini-nanobababa2

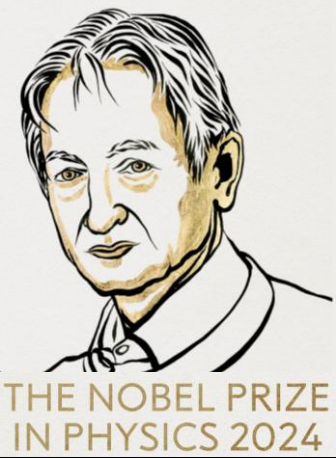
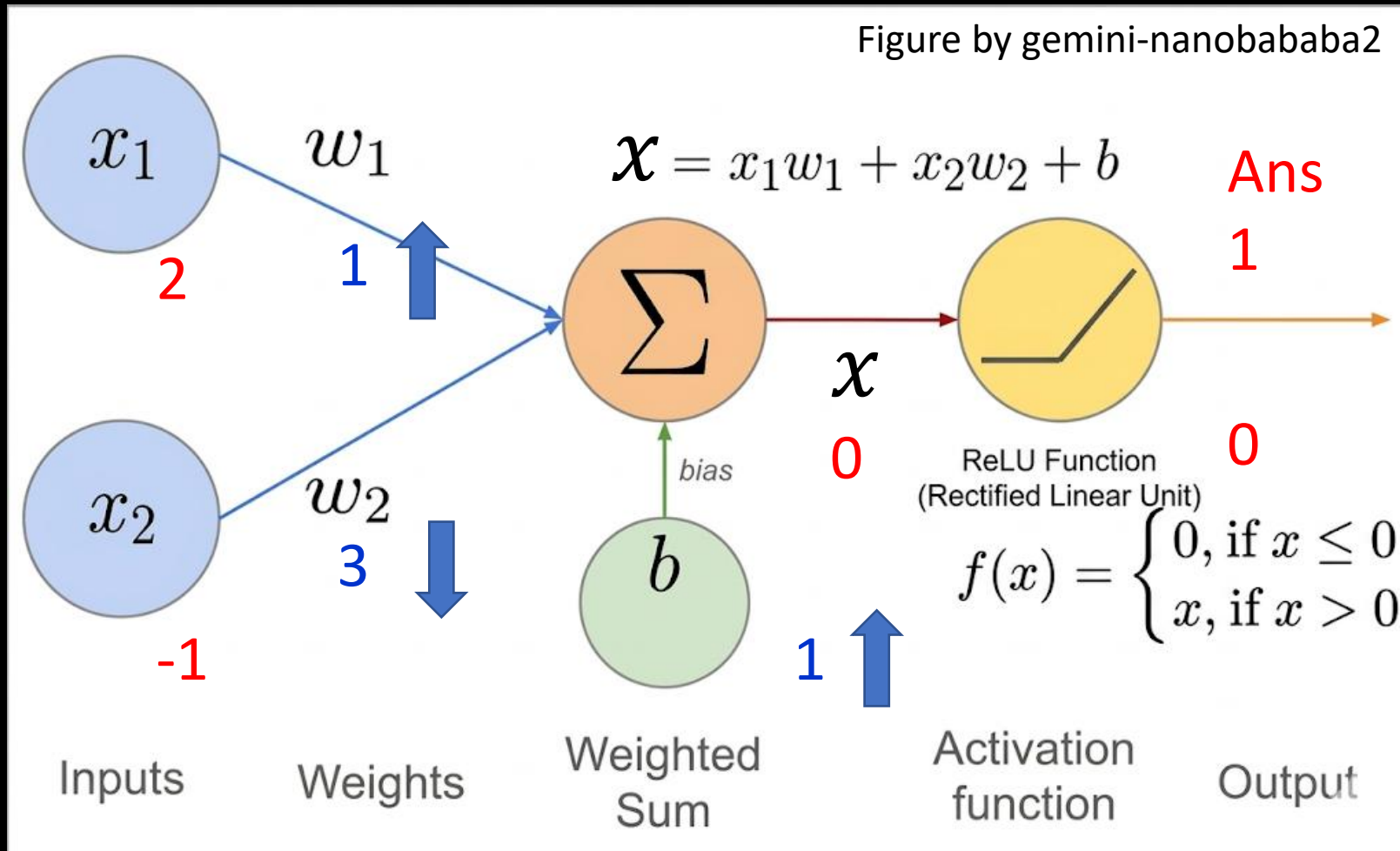


<https://en.wikipedia.org/wiki/Neuron>

- w_1, w_2, b are parameters
- Learning, is finding the best parameter

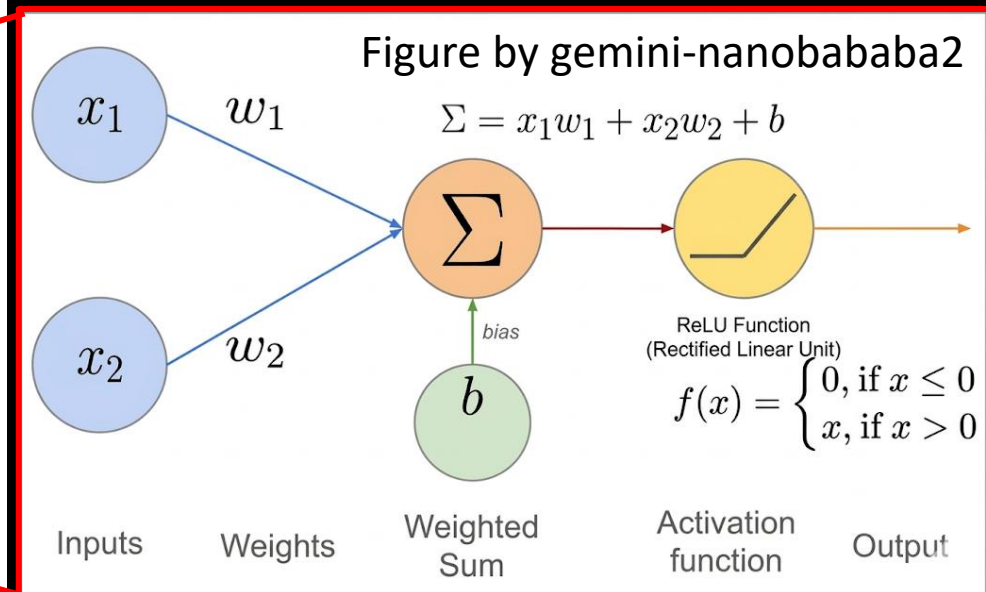
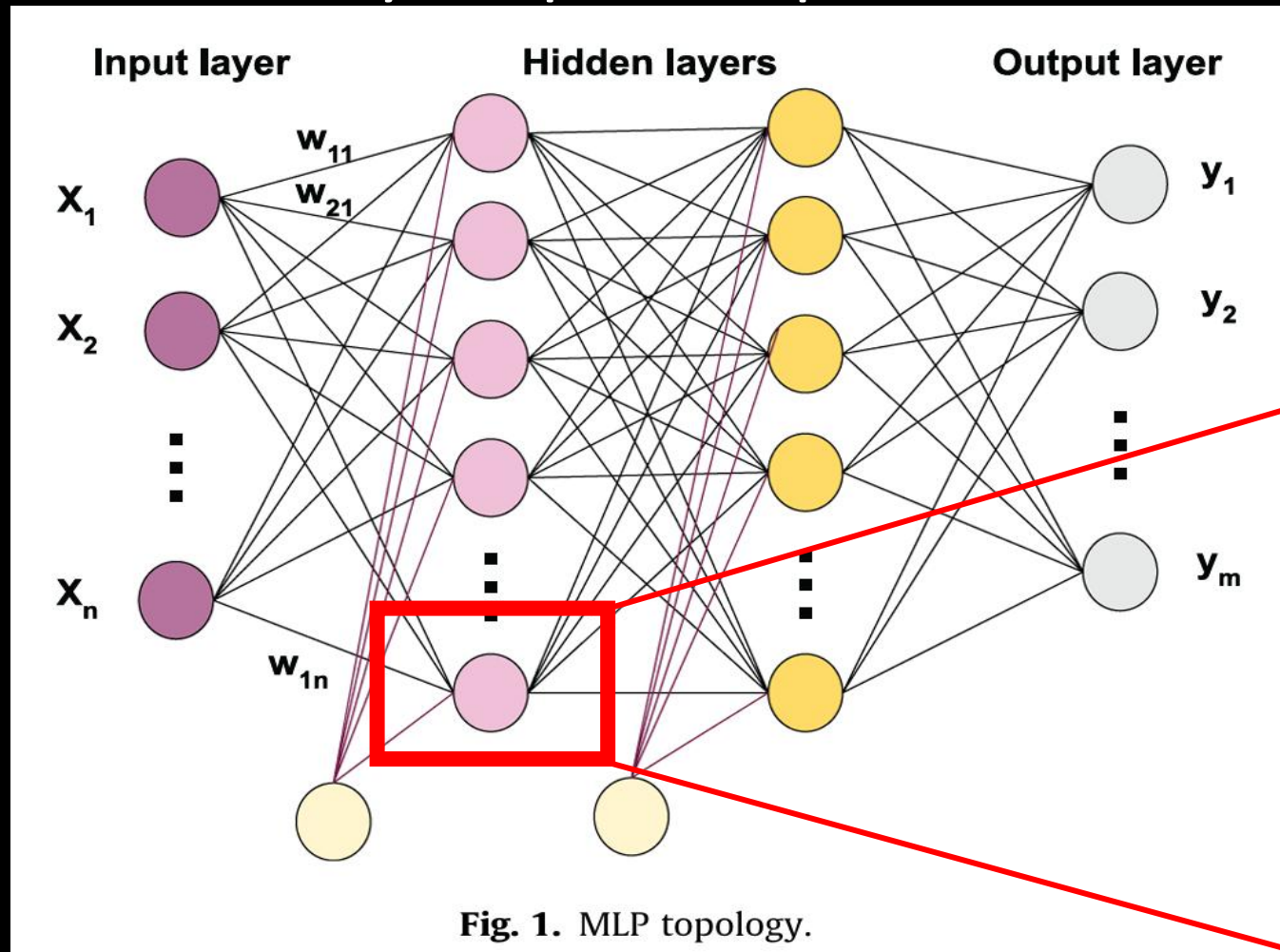
How to find better parameter?

- $$\frac{-d \text{Error}}{d w_1} = \frac{-d \text{Error}}{d f(x)} \frac{d f(x)}{d x} \frac{\partial x}{\partial w_1} = -x_1$$



Back propagation

How to learn more things? multi-layer perceptron



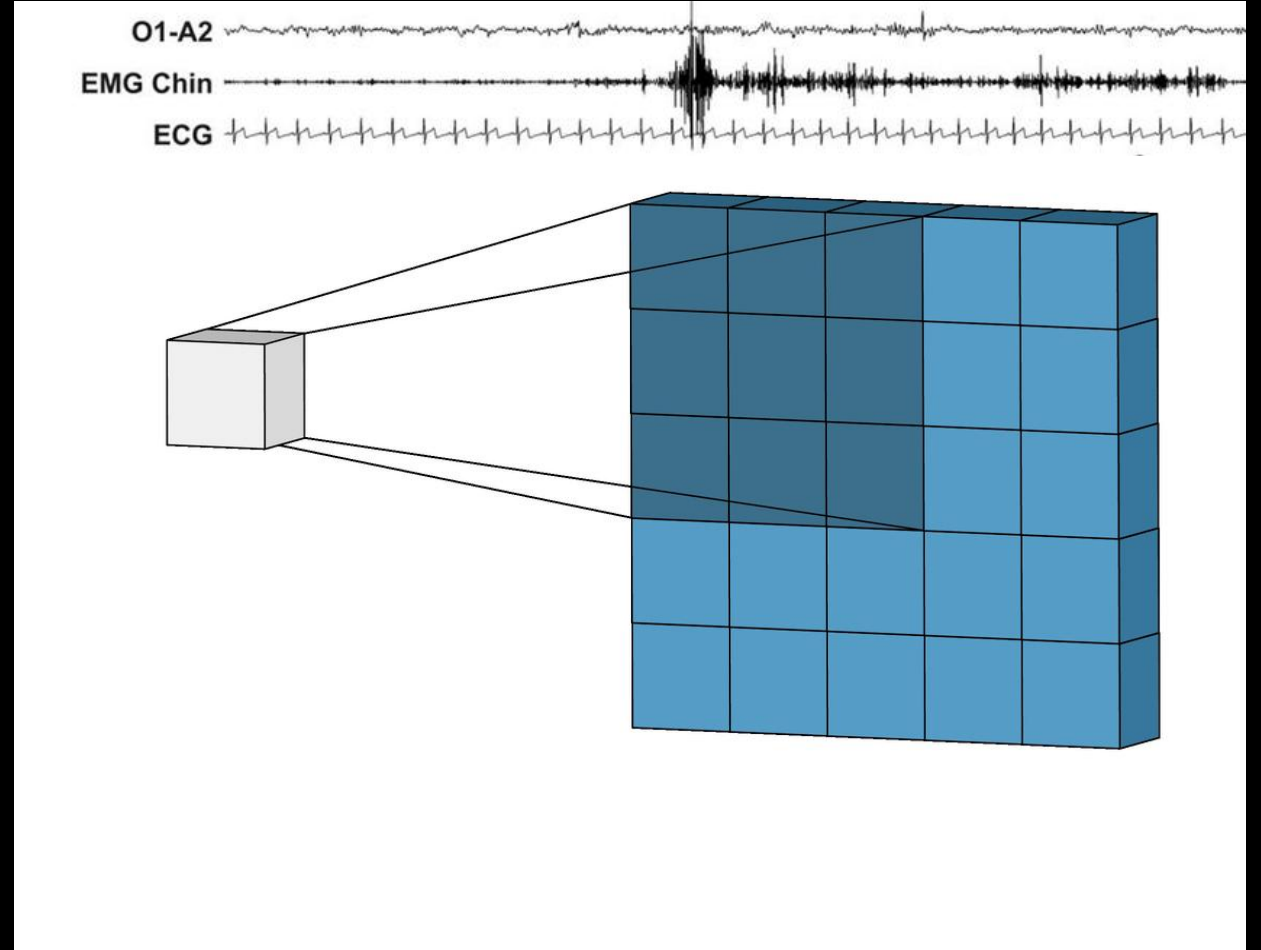
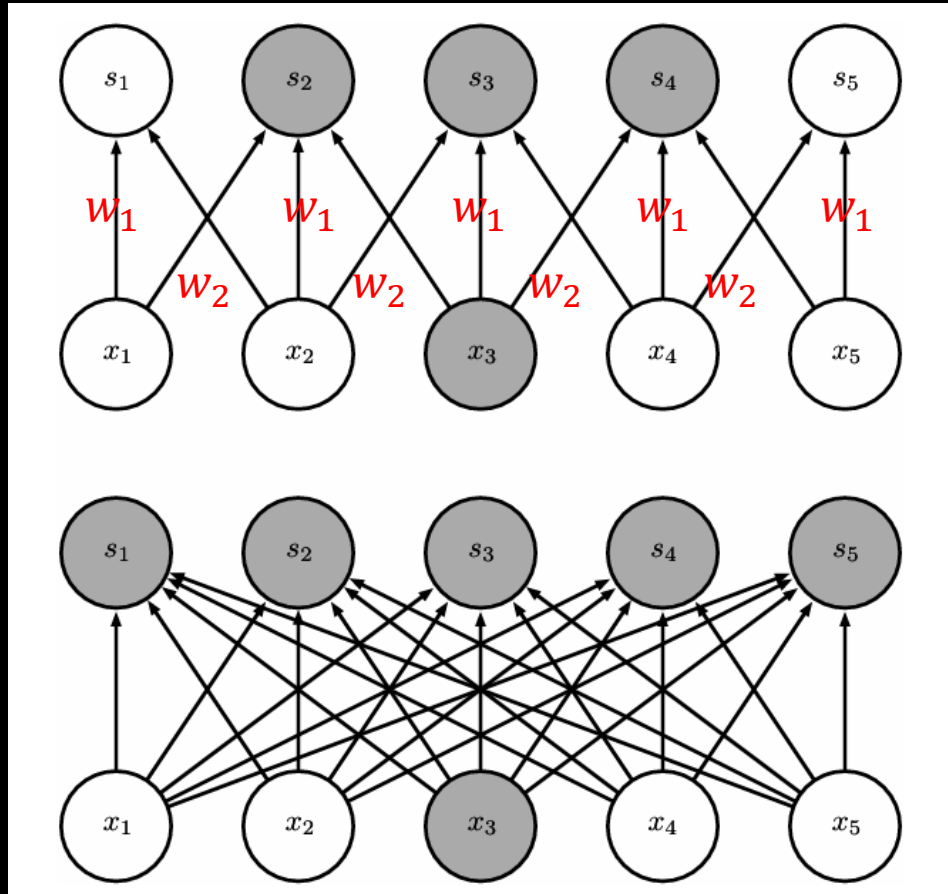
What does it cost to make it very large?

- $x_n: (x_{n1}, x_{n2}, x_{n3}, x_{n4})$ is a patient with 4 features
- $w_1x_{11} + w_2x_{12} + w_3x_{13} + w_4x_{14} = b_1$
- $w_1x_{21} + w_2x_{22} + w_3x_{23} + w_4x_{24} = b_2$
- $w_1x_{41} + w_2x_{42} + w_3x_{43} + w_4x_{44} = b_4$

In fact, neural network is not linear.

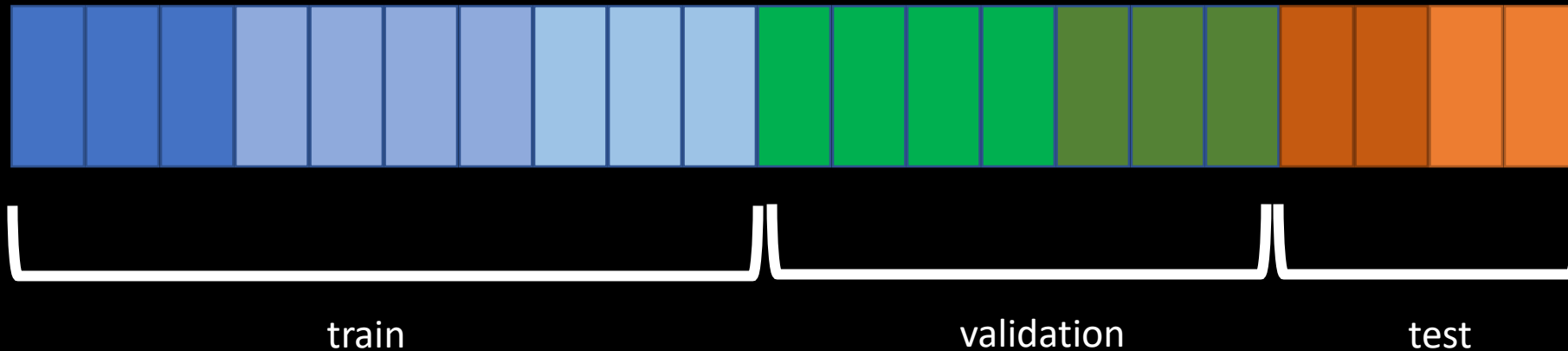
This is only for a taste and having intuition.

How to reduce parameter? convolution



Data preparation

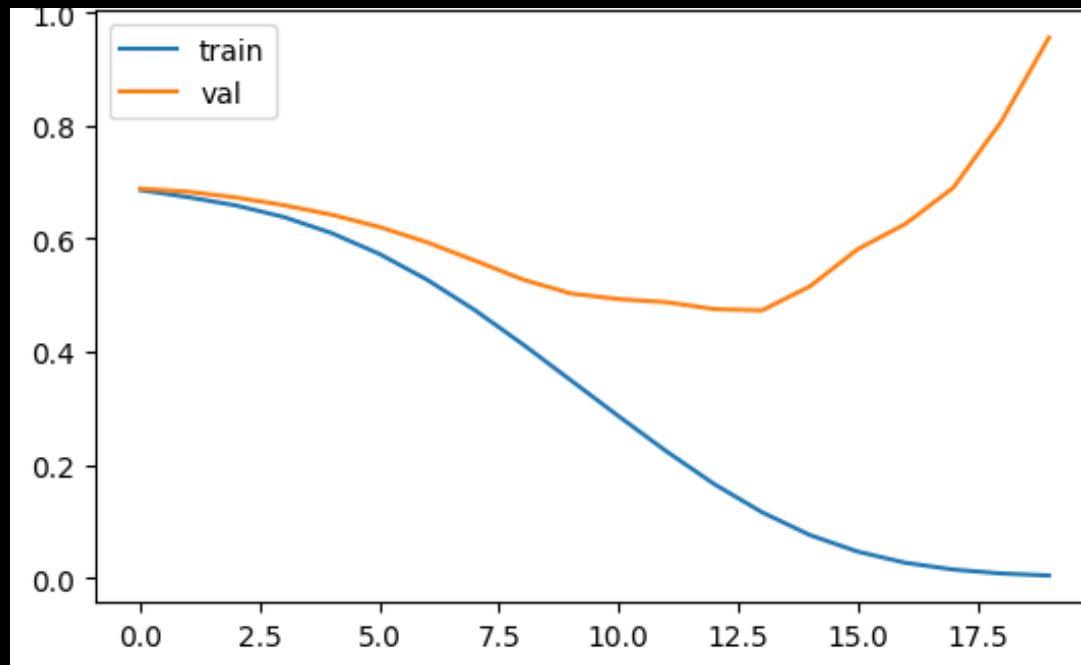
- Patient based separation
- Train: update parameter
- Validation: inform and adjust the training process
- Test: evaluate the final performance



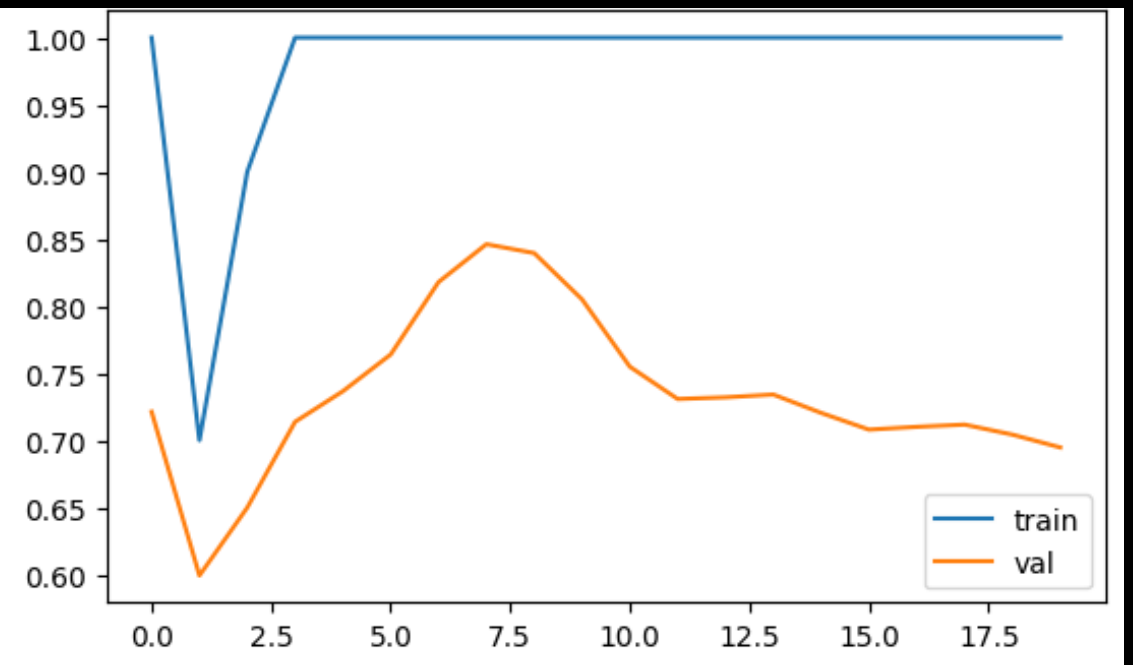
Overfit

We remember the test question, but not understand the knowledge

Error



Accuracy

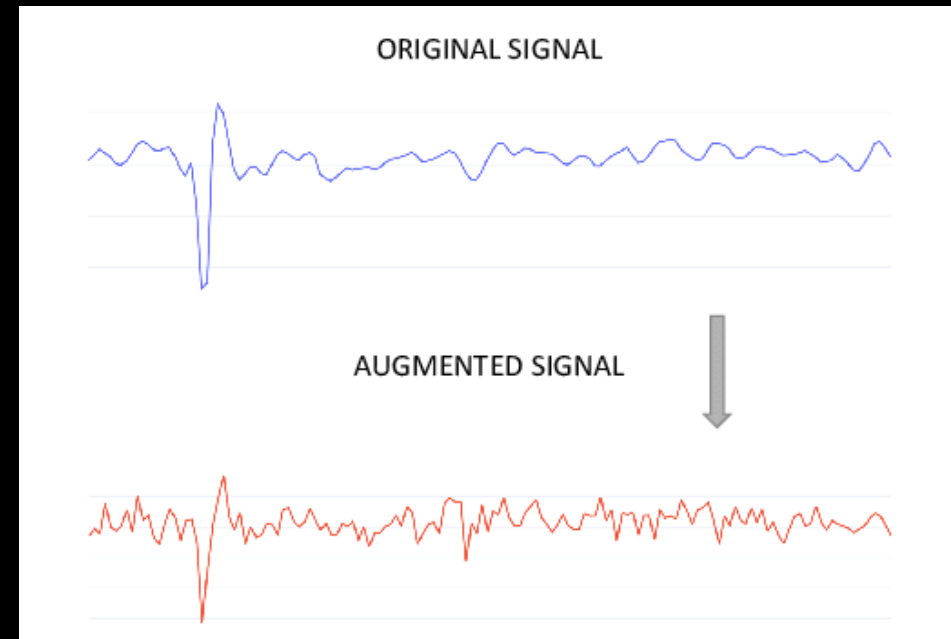


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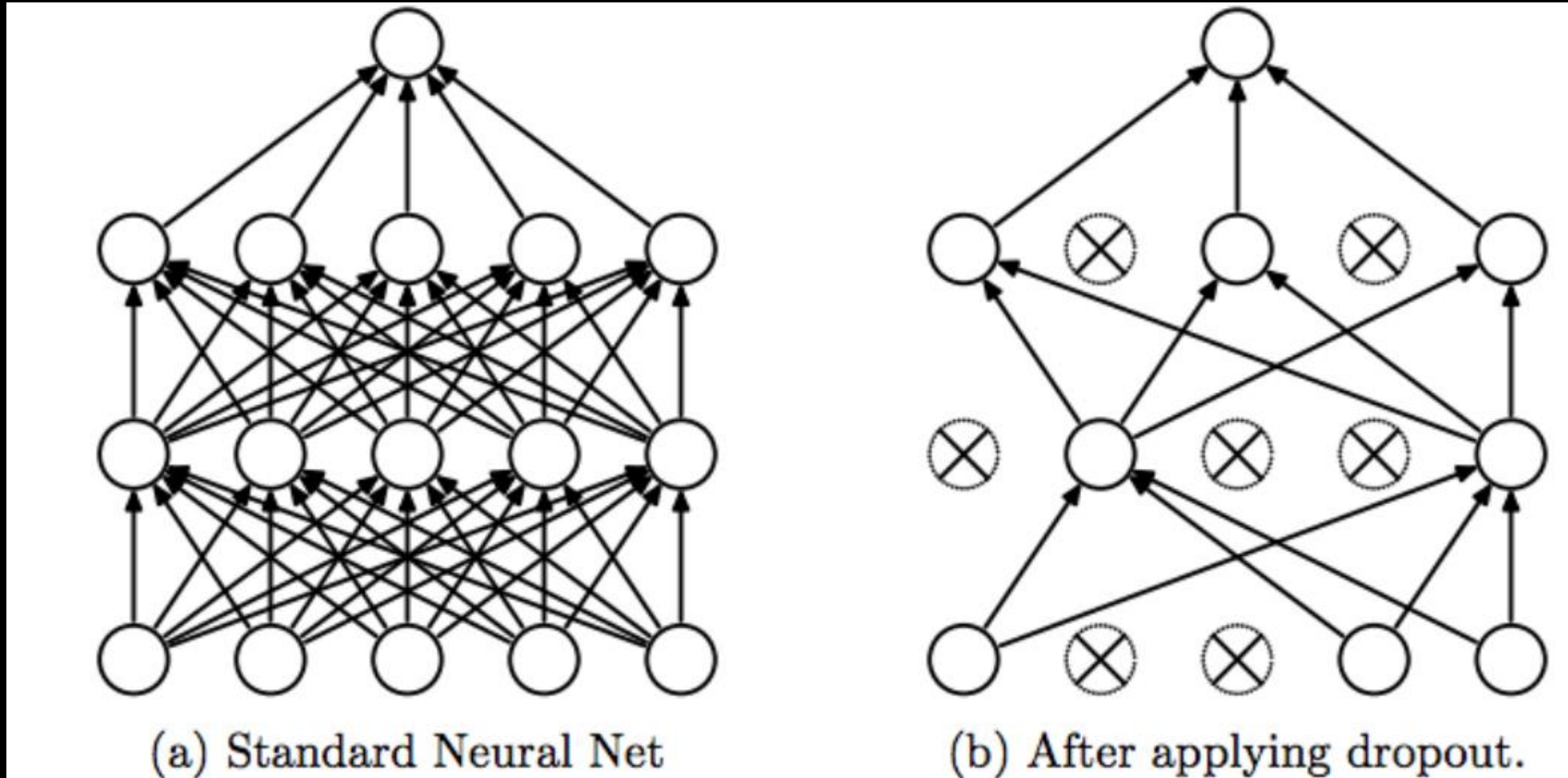
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How to prevent memorizing answers

- Fundamental solution: more questions in question bank
- Data augmentation: change question a little bit each time.
 - Add noise
 - Time shift
 - Phase angle rotation
 - ...



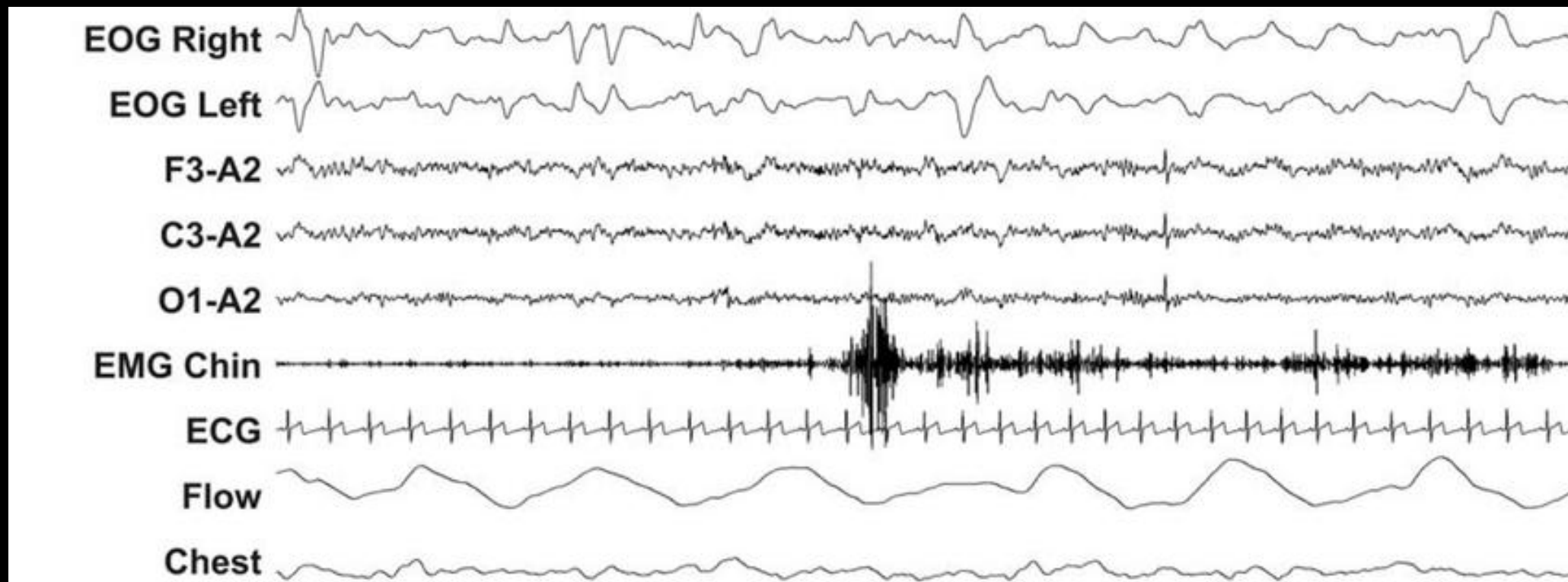
Dropout



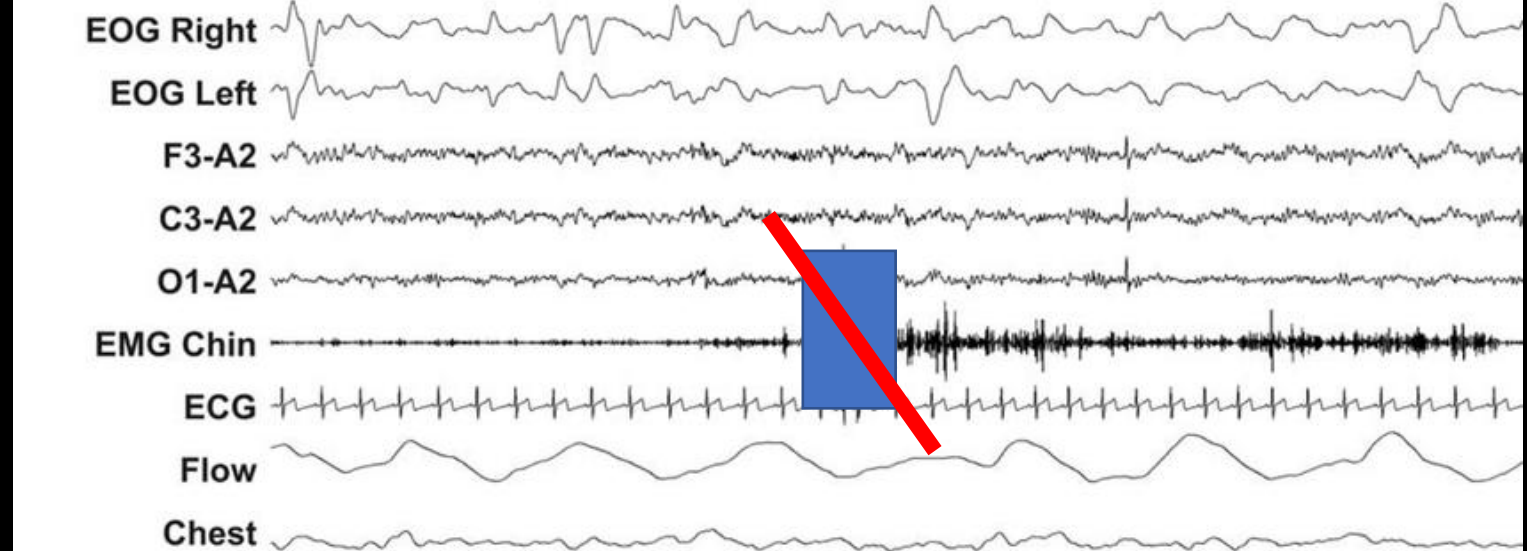
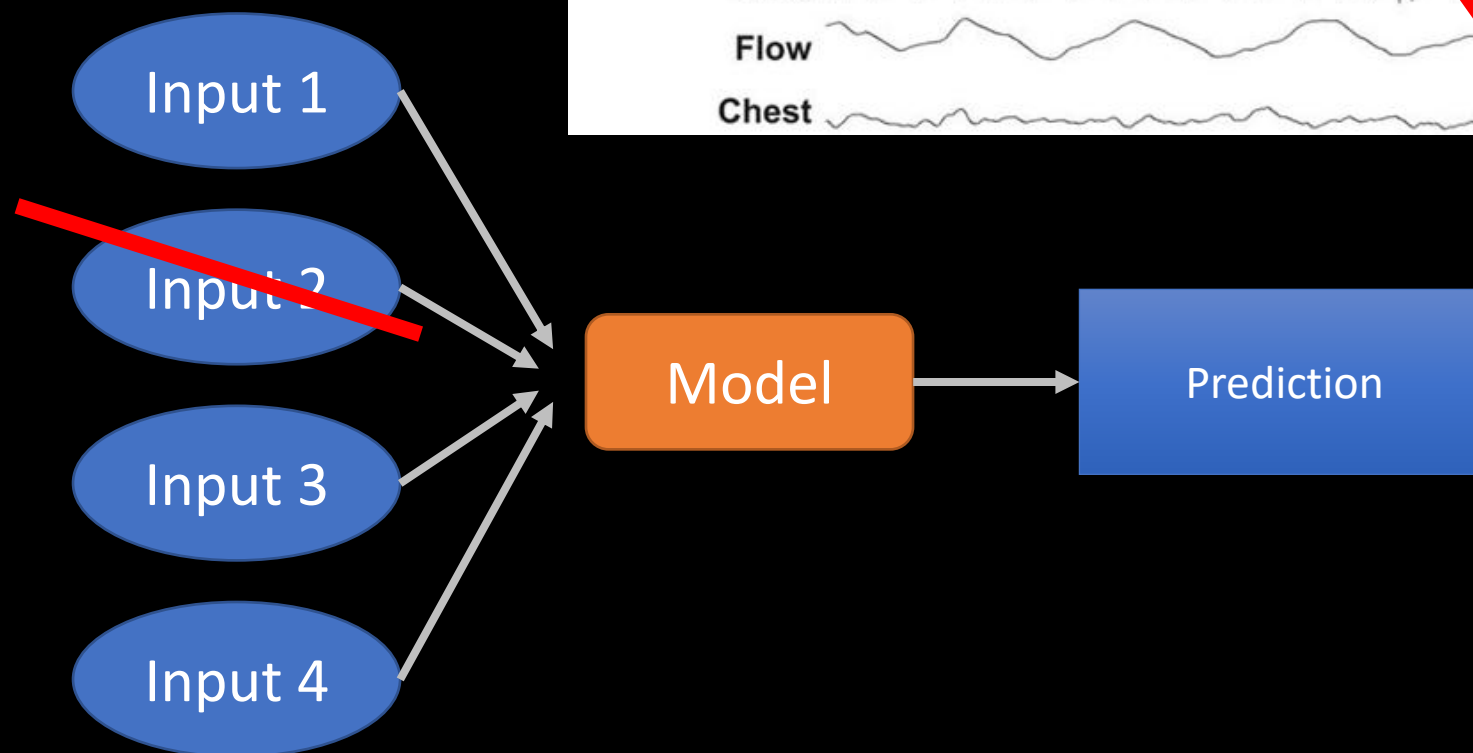
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Which part of the input affect model prediction?



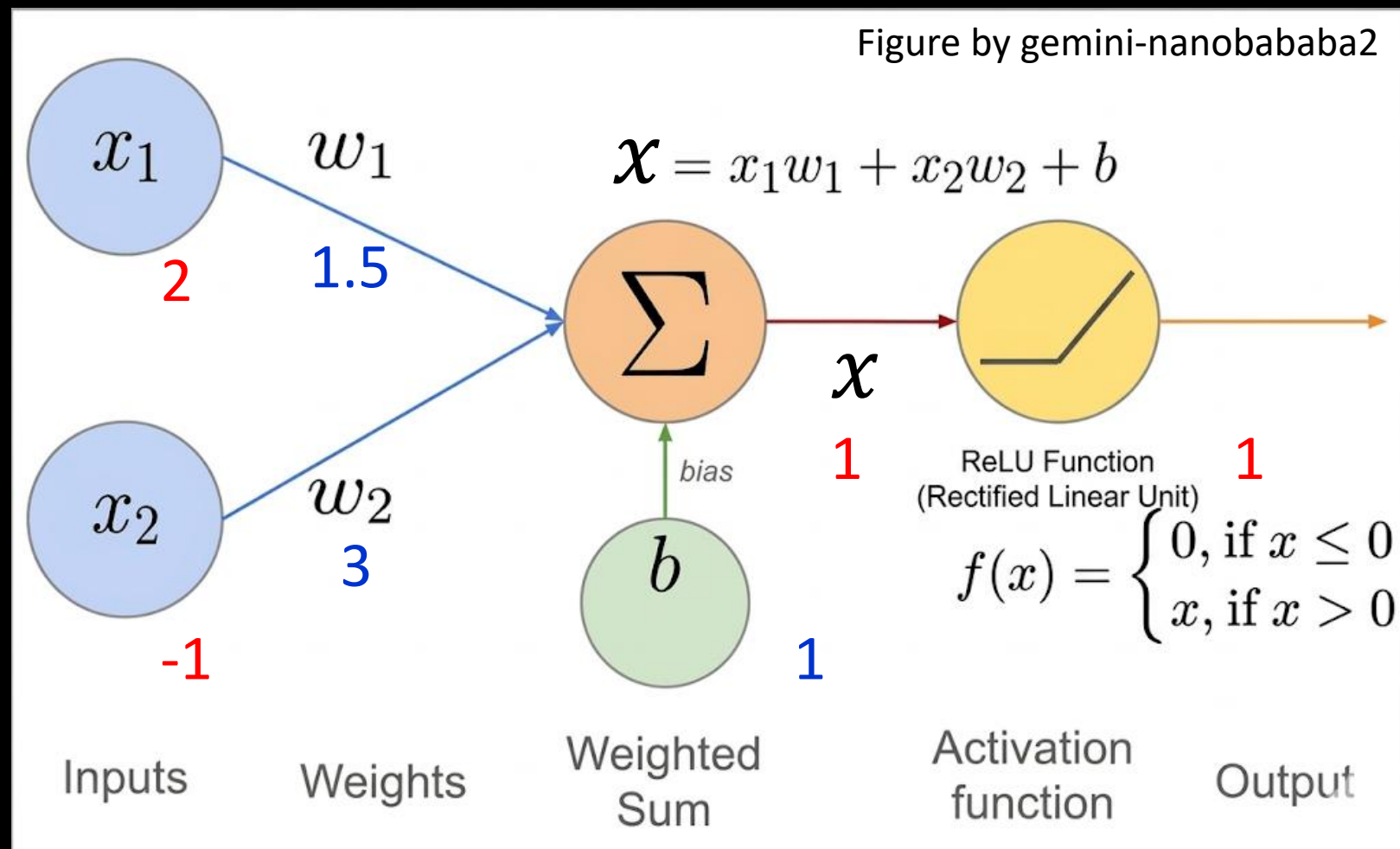
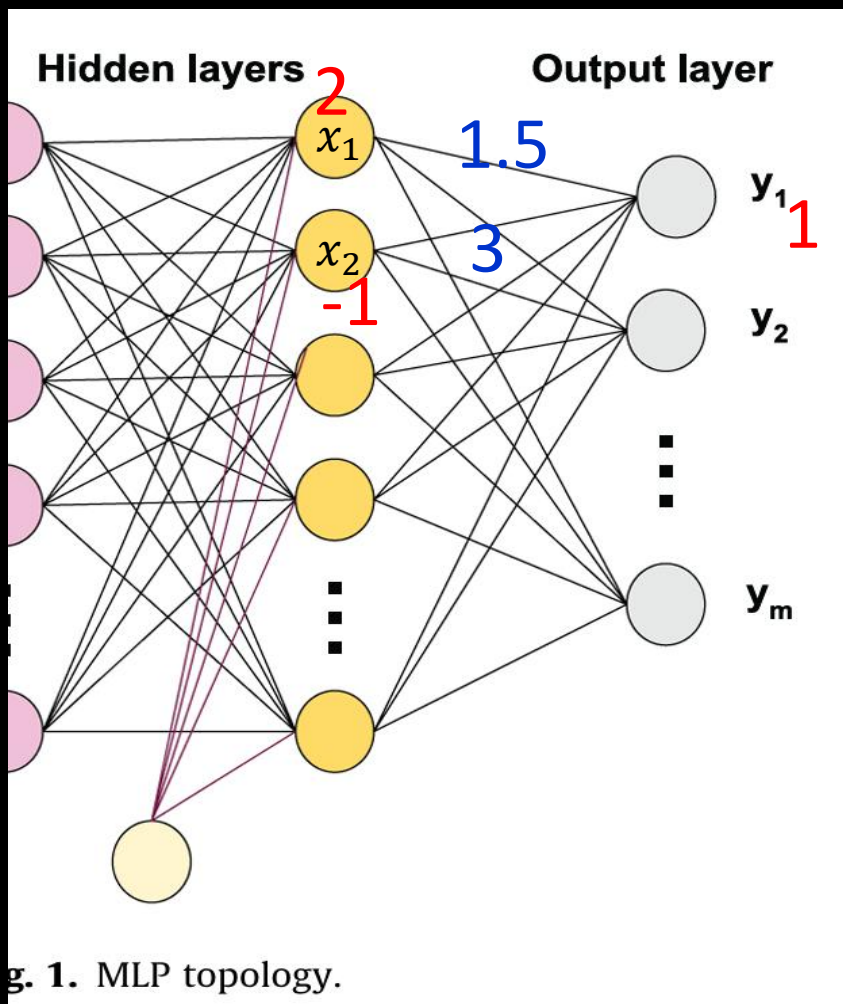
Ablation



How hard is x_1 and x_2 push the output toward current "1"?

Grad-Cam

$$Push_{x_1} = x_1 \frac{d \text{Output}}{d x_1} \quad F=ma$$



Deep learning in biomedical data

- Catch implicit relationship between data and outcome.
 - Sometimes, we only have label, but not clear cut criteria.
 - High dimension, non-linear, less feature engineering.
- Why doing it in REM sleep with atonia?
 - Sometimes, we didn't have all channels.
- The key beyond engineering is always “more high quality data”.



[https://github.com
/Po-yuLin/AOC-
IFCN_workshop](https://github.com/Po-yuLin/AOC-IFCN_workshop)



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